Meta-Learning Guided Weakly Supervised Video Anomaly Detection with Dual Memory and Temporal Attention

Abstract

Weakly supervised anomaly detection (WSAD) aims to automatically identify abnormal events in surveillance videos since obtaining precise frame-level annotations is expensive and often ambiguous. However, feature ambiguity, overfitting to dominant normal patterns, and poor adaptability to ambiguous samples are often encountered in WSAD. In this work, we proposed a model agnostic meta-learning (MAML) framework for weakly supervised video anomaly detection that enhances generalization across heterogeneous anomaly types using video-level supervision. We used a dual memory architecture that separately models normal and abnormal prototypes, facilitating effective disentanglement of sparse anomalous cues from dominant normal patterns. Transformer based attention, multiple instance learning (MIL), metric learning and latent uncertainty regularization further enforce robust separation between normality and anomaly, improving temporal localization performance. We achieved an AUC of 86% and an accuracy of 87%, outperforming several recent weakly supervised baselines.

Keywords: WSAD, MAML, Dual memory, Top-K MIL, Uncertainity Regulation, UCF-Crime

1 Introduction

Anomaly detection in long, untrimmed videos remains one of the most challenging problems in computer vision. Real-world surveillance footage exhibits large variability in appearance, motion patterns, temporal duration, and scene context, while anomalous events are inherently rare and weakly defined. As a result, collecting dense frame-level annotations is expensive and often impractical, rendering fully supervised learning approaches unsuitable for large-scale deployment.

To address this challenge, Weakly Supervised Anomaly Detection (WSAD) has emerged as a promising paradigm, where models are trained using only video-level labels that indicate whether an anomaly occurs somewhere within a video. But despite significant progress, WSAD based models fail to localize anomalous segments without explicit temporal supervision.

Most existing WSAD approaches adopt Multiple Instance Learning (MIL) combined with attention mechanisms to infer anomalous segments. Sultani et al. (2018); Tian et al. (2021); Kanishkar et al. (2024) Neverthless, these methods face challenges like poor generalization to unseen anomaly types, unstable temporal attention, and limited adaptability when only a few examples of new anomalies are available Wu et al. (2022). These limitations arise because conventional WSAD models are trained using a single global objective and lack mechanisms to explicitly model task-level variability Ilse et al. (2018).

To overcome these challenges, we propose a Model-Agnostic Meta-Learning (MAML) framework for WSAD as it enables rapid adaptation of both decision boundaries and temporal attention using only a few weakly labeled examples. Each training iteration is treated as a distinct task composed of a support set and a query set, forcing the model to learn representations that generalize across heterogeneous anomaly distributions.

To further enhance discriminative power, we integrated an enhanced WSAD architecture equipped with transformer-based temporal modeling, dual memory banks for normal and abnormal patterns, metric-learning constraints, variational latent regularization, and temporal smoothness losses. This hybrid design unifies memory-augmented representation learning with meta-optimization, yielding a robust and adaptive video anomaly detection system.

Video anomaly detection (VAD) aims to identify events or behaviours that deviate from normal patterns in untrimmed video streams. In surveillance scenarios, anomalies typically correspond to safety-critical incidents where automated detection can significantly reduce human monitoring effort and enable timely intervention Sultani et al. (2018); Roy et al. (2018); Direkoglu (2020); Amrutha et al. (2020); Viji et al. (2024). Despite extensive research, VAD remains challenging due to several intrinsic factors. Anomalous events are inherently rare, diverse, and context-dependent, requiring detection systems to recognise a wide range of abnormal behaviours while maintaining low false positive rates when benign activities resemble anomalies. Real-world surveillance videos often suffer from low resolution, illumination variations, occlusions, and pose variations, complicating robust feature extraction Gandapur and Verdú (2023). Anomalies are also temporally sparse and short-lived, necessitating effective temporal modelling for precise localisation. These challenges are exacerbated by limited availability of fine-grained annotations, especially in densely populated areas motivating weakly supervised learning approaches Sultani et al. (2018); Zhang et al. (2022).

Early approaches to VAD primarily relied on handcrafted features or motion-based representations.Direkoglu (2020) employed motion information images combined with convolutional neural networks to detect abnormal crowd behaviour, achieving promising results in dense scenes but struggling with short, localised anomalies. Roy et al. (2018) proposed an action-attribute-based framework for snatch-theft detection, capturing fine-grained motion cues such as abrupt arm movements and victim proximity; however, the approach was sensitive to occlusions and viewpoint changes. Hybrid architectures combining convolutional networks with recurrent models, such as ConvGRU CNN frameworks, were later explored to model spatiotemporal dependencies more

effectively Gandapur and Verdú (2023), though their ability to capture long-range temporal context remained limited.

UCF-Crime dataset is popularly used for weakly supervised anomaly detection an one of the first works that used this dataset fro anomaly detection as Sultani et al. (2018). This work formalised VAD under the MIL paradigm, where each video is treated as a bag of temporal snippets, and anomalous videos are assumed to contain at least one abnormal instance. Subsequent MIL-based methods focused on improving instance selection and feature representation. Authors Tian et al. (2021) proposed robust temporal feature magnitude learning (RTFM) to better distinguish abnormal snippets.While Feng et al. (2021)introduced a self-training strategy generating pseudo labels to progressively refine anomalous regions. Both approaches emphasised learning discriminative anomalous features but often neglected explicit modelling of normality, leading to overfitting. Zhang et al. (2022) extended MIL frameworks by considering the openness of events, addressing unseen anomalies in surveillance data. Wu et al. (2021) employed weakly supervised spatiotemporal modelling to improve anomaly localisation in challenging environments.

Memory-augmented architectures have been explored to explicitly model normal patterns. Gong et al. (2019) proposed MemAE, a memory-augmented autoencoder that memorises normal prototypes and suppresses anomaly reconstruction. Stephen and Menon (2020) introduced a readable and updatable memory scheme, allowing memory updates at test time. Liu et al. (2021) designed multi-level memory modules in autoencoders with skip connections to memorise normal patterns for optical flow reconstruction. Luo et al. (2021) demonstrated the effectiveness of action unit memory networks in weakly supervised temporal localisation, motivating the extension of memory mechanisms to anomaly detection. More recently, uncertainty-regulated dual memory units (Zhou et al. (2023)) have been proposed to jointly model normal and anomalous patterns, enhancing representation and generalisation Zhou et al. (2023).

Data uncertainty learning has also emerged as an effective strategy to handle noisy and ambiguous supervision. Chang et al. (2020) proposed data uncertainty modelling to learn feature embeddings with associated uncertainty in face recognition. Lee et al. (2021) applied uncertainty learning in temporal action localisation by treating background frames as out-of-distribution (OOD) samples.

Transformer-based architectures have recently gained attention in VAD due to their ability to capture long-range temporal dependencies and global context. Vision transformers and spatiotemporal self-attention mechanisms have demonstrated strong performance in modelling both local and global interactions Dosovitskiy et al. (2021); Liu et al. (2021); Vaswani et al. (2017). Habeb et al. (2024) proposed an unsupervised transformer framework with spatiotemporal relation attention blocks, achieving improved anomaly localisation in large-scale datasets. Shao et al. (2023) introduced CoVAD, leveraging self-attention for enhanced spatial and temporal anomaly detection. Dilek and Dener (2025) surveyed transformer-based VAD methods, highlighting their effectiveness in sparse and context-dependent anomaly scenarios while noting the computational costs. Bertasius et al. (2021) explored space-time attention transformers for video understanding, further motivating the integration of attention mechanisms in VAD. Dosovitskiy et al. Dosovitskiy et al. (2021) and Liu et al. Liu et al. (2021)

demonstrated the advantages of hierarchical and patch-based transformer designs for capturing global context.

The exhaustive studies demonstrated the evolution of VAD from motion and CNN-centric methods to attention based and memory-augmented architectures. It highlights persistent challenges including rare unseen anomalies, data imbalance and poor generalisation. These gaps leads to the development of hybrid approaches that leverage MIL based supervision, dual memory architectures, uncertainty-regulated latent representations, and attention-based temporal modelling mechanism to achieve robust and generalisable video anomaly detection system.

In our extensive literature study, we identified several research gaps and challenges that persist in video-based anomaly detection, particularly for crime scenarios. A major limitation is the scarcity of large-scale, publicly available open source labeled surveillance datasets especially region or geo specific.Existing methodologies often rely on synthetic or artificially curated datasets, which cannot mimic real-world scenarios hence the prediction struggles in generalization over unseen and rare events. While models trained on global datasets performs poorly in adapting to Indian environments specifically due to cultural, behavioural, and infrastructural differences.CCTV footages captured in degraded atmospheric conditions, poor illumination, occlusions, complex crowd behavior where distinguishing victim's and perpetrators behavior becomes challenging, often results in deteriorating performance. Behavioural ambiguity also plays a significant role in raising high false alarm rates, as actions such as running or sudden movement may be contextually normal yet misclassified as anomalous, and subtle, short-duration events are harder to detect than apparent violent incidents.

From a methodological perspective, many approaches fail to achieve precise temporal localisation of anomalies within long video sequences and inadequately model long-term spatiotemporal dependencies, limiting fine-grained recognition. Furthermore, algorithmic gaps remain, including limited exploration of meta-learning for few-shot adaptation to unseen crimes, insufficient integration of transformer and autoencoder architectures for robust spatiotemporal modelling, poor generalisation to open-set anomalies, and instability arising from randomized sampling strategies in meta-learning frameworks.

To improve generalisation across diverse environments and unseen crime patterns, we introduced a meta-learning-driven framework that enables rapid adaptation to heterogeneous anomaly types with limited supervision. By learning task-invariant representations, the proposed model alleviates overfitting to dominant normal behaviours and improves robustness to cultural, behavioural, and infrastructural variations commonly observed in Indian surveillance settings. A dual memory architecture is incorporated to explicitly disentangle normal and abnormal prototypes, reducing behavioural ambiguity and lowering false alarm rates in crowded and visually complex scenes.to overcome methodological limitations in temporal modelling and localisation, we employ a transformer-based temporal encoder augmented with local attention, enabling effective modelling of both long-range temporal dependencies and

short-duration, subtle events such as chain-snatching. The integration of metric learning, latent uncertainty regularisation, and temporal smoothness constraints further stabilises training and enhances precise temporal localisation under weak supervision

2 Proposed Methodology

In WSAD, each training video is annotated only with a video-level label indicating whether it contains anomalous events, while frame-level anomaly annotations are unavailable. Our proposed technique is augmented with MAML which has been demonstrated (Zhang et al. (2022); Naik et al. (2025)) to enhance anomalous event detection in videos.

Weakly supervised video anomaly detection suffers from severe task distribution shifts, where the types, durations, and visual manifestations of anomalies observed during training differ significantly from those encountered at test time. Traditional WSAD methods implicitly assume that anomaly representations learned from limited training categories generalize to unseen anomalies, an assumption that often fails in real-world surveillance scenarios.

To address this limitation, we introduced a meta-learning strategy that explicitly optimizes the model for rapid adaptation across tasks. Each video (or batch of videos) is treated as a task, where high-confidence snippets selected via multiple instance learning form a support set, and the remaining snippets constitute a query set. The model is trained such that a small number of gradient updates on the support set yields improved anomaly localization on the query set.

By optimizing task-level generalization rather than static anomaly recognition, meta-learning complements the proposed dual memory architecture and uncertainty-aware learning framework, enabling robust detection of unseen and temporally sparse anomalies.

Each video is uniformly divided into T temporal snippets, and each snippet is represented by a pretrained feature vector. We used I3D RGB features pre-trained on Kinetics dataset, where 16 consecutive frames were used for a single feature. Under the (MIL) assumption Wang et al. (2018); Ilse et al. (2018), a normal video contains no anomalous snippets, whereas an abnormal video contains at least one anomalous snippet.

Our objective is to learn a model that produces temporally precise anomaly scores while being trained using only weak supervision.

2.1 Architecture

Inspired by Zhou et al. (2023), our proposed framework integrates temporal modeling, memory-based representation learning, metric regularization, and meta-learning optimization with MAML, as shown in Figure 1. Given an input video feature sequence

$$\mathbf{X} \in \mathbb{R}^{T \times D}, \tag{1}$$

T and D being the temporal snippets and dimension respectively, the model processes the input through the following stages:

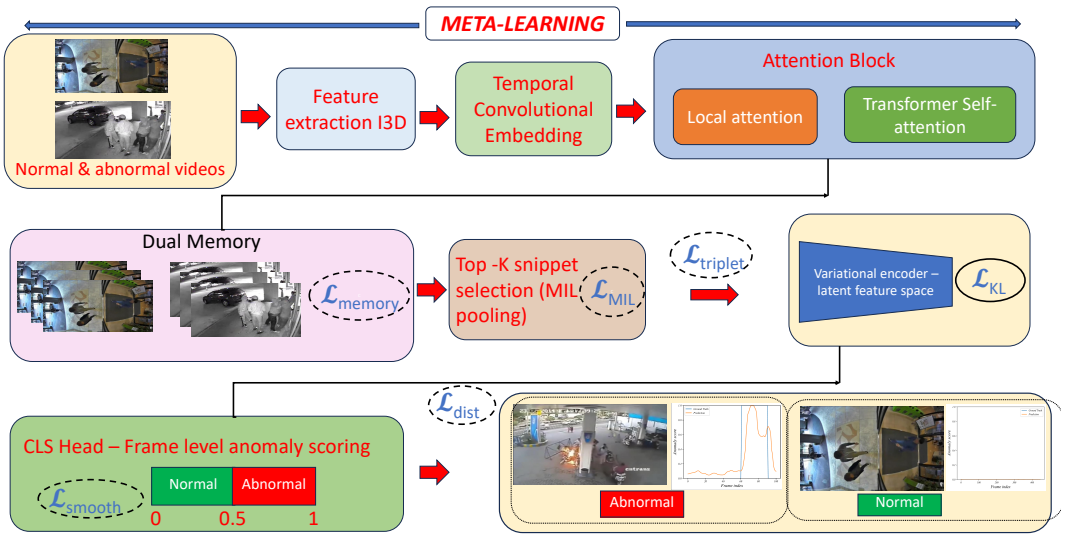


Fig. 1: Architecture of the proposed MAML-based video anomaly detection framework

- Temporal convolutional embedding
- Local temporal attention
- Global self-attention via Transformer
- Dual memory units (normal and abnormal)
- Memory-augmented classification head

The overall design aims to capture both short-term and long-range temporal dependencies while explicitly separating normal and abnormal patterns in feature space. All of these stages are described in this section.

To capture local temporal continuity and reduce feature dimensionality, we first applied a 1D temporal convolution:

$$\mathbf{X}_e = \text{Conv1D}(\mathbf{X}), \tag{2}$$

where $\mathbf{X}_e \in \mathbb{R}^{T \times d}$ and $d \ll D$. This layer mainly works mainly to reduce input dimensions to aid the subsequent steps. This also aids in smoothing the input.

A typical anomalous video has several normal frames and only a small number of abnormal frames. Hence we introduce a local attention mechanism, represented as LocalConv implemented with temporal depthwise convolution with a small kernel size. This means that we perform convolution initially over each dimension separately, across time frames. We use a kernel size of 9. This is followed by a pointwise convolution, which aggregates information across features:

$$\mathbf{X}_l = \mathbf{X}_e + \text{LocalConv}(\mathbf{X}_e). \tag{3}$$

where

$$\text{LocalConv}(\mathbf{X}_e) = \text{Conv}_{1\times1}\left(\sigma\left(\text{DWConv}_k(\mathbf{X}_e)\right)\right)$$

and

$$[\text{DWConv}_k(\mathbf{X})]_{t,c} = \sum_{i=-\lfloor k/2 \rfloor}^{\lfloor k/2 \rfloor} w_{c,i}\, X_{t+i,c}$$

$$[\text{Conv}_{1\times1}(\mathbf{X})]_{t,c} = \sum_{c'=1}^{d} W_{c,c'}\, X_{t,c'}$$

Where $w_{c,i}$ is the convolution weight for channel c and time offset i, $W_{c,c'}$ are learnable parameters which are determined during training. $k = 9$ in our model. Though the model does include global attention subsequently, addition of temporal attention smoothing is a unique aspect of our work and significantly improves MAML performance, since it smooths noisy gradients from meta-learning steps.

To capture long-range temporal dependencies and contextual relationships across snippets, we employed a Transformer-based self-attention encoder:

$$\mathbf{X}_g = \text{Transformer}(\mathbf{X}_l). \tag{4}$$

This allows the model to reason globally over the video, distinguishing isolated noise from consistent abnormal behavior.

We maintain two learnable memory banks:

- Normal memory $\mathcal{M}_n$,
- Abnormal memory $\mathcal{M}_a$,

each consisting of M prototype vectors. Given a feature sequence $\mathbf{X}_g$, we compute memory query scores and memory-augmented features:

$$\mathbf{S} = \sigma\left(\frac{\mathbf{X}_g\mathbf{M}^\top}{\sqrt{d}}\right), \quad \mathbf{X}_{\text{aug}} = \mathbf{S}\mathbf{M}, \tag{5}$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\mathbf{M} \in \mathbb{R}^{M\times d}$ is a learnable memory bank matrix. Due to the above operations, the final matrix $\mathbf{X}_{\text{aug}} \in \mathbb{R}^{T\times d}$. In the calculation of $\mathbf{S}$, $\sqrt{d}$ is required in the denominator to prevent large dot product magnitudes. It follows the same principle as in the scaled-dot product attention formula Vaswani et al. (2017).

To stabilize memory learning under weak supervision, we performed a second memory read on the augmented features:

$$\tilde{\mathbf{X}}_{\text{aug}} = \mathbf{X}_{\text{aug}} + \frac{1}{2}\,\sigma\left(\frac{\mathbf{X}_{\text{aug}}\mathbf{M}^\top}{\sqrt{d}}\right)\mathbf{M} \tag{6}$$

This reinforcement suppresses noisy activations and strengthens consistent temporal patterns.

To explicitly disentangle normal and abnormal patterns, the framework employs dual memory units that produce attention responses:

$$\mathbf{A}_{att},\ \mathbf{N}_{att},\ \mathbf{A}\to\mathbf{N}_{att},\ \mathbf{N}\to\mathbf{A}_{att}, \tag{7}$$

corresponding to abnormal memory activation, normal memory activation, abnormal-to-normal memory read, and normal-to-abnormal memory read, respectively. Basically, $\mathbf{A}_{att}$ and $\mathbf{N}_{att}$ are the responses from $\mathcal{M}_a$ and $\mathcal{M}_n$ respectively. $\mathbf{A}\to\mathbf{N}_{att}$ and $\mathbf{N}\to\mathbf{A}_{att}$ are the cross-responses when abnormal and normal samples are passed through $\mathcal{M}_n$ and $\mathcal{M}_a$ respectively. The model learns to suppress cross-activations and encourages self-activation, i.e. normal and abnormal snippets reinforce their own kind, while suppressing the other.

Abnormal Memory Activation Loss

For abnormal videos, the abnormal memory is encouraged to strongly activate on anomalous snippets with a Binary Cross-Entropy (BCE):

$$\mathcal{L}_A = \mathrm{BCE}\left(\frac{1}{K}\sum_{t\in\mathrm{TopK}(\mathbf{A}_{att})} A_{att}^{(t)},\ 1\right). \tag{8}$$

Normal Memory Consistency Loss.

Normal memory activations are encouraged to remain high for normal patterns:

$$\mathcal{L}_N = \mathrm{BCE}(\mathbf{N}_{att}, \mathbf{1}). \tag{9}$$

Cross-Memory Suppression Losses

To prevent memory confusion, asymmetric cross-memory constraints are imposed:

$$\mathcal{L}_{A\to N} = \mathrm{BCE}(\mathbf{A}\to\mathbf{N}_{att}, \mathbf{0}), \tag{10}$$

$$\mathcal{L}_{N\to A} = \mathrm{BCE}\left(\frac{1}{K}\sum_{t\in\mathrm{TopK}(1-\mathbf{N}\to\mathbf{A}_{att})} (1-N\to A_{att}^{(t)}),\ 1\right). \tag{11}$$

Overall Memory Loss

The total memory regularization term is defined as:

$$\mathcal{L}_{\text{memory}} = \mathcal{L}_A + \mathcal{L}_N + \mathcal{L}_{A\to N} + \mathcal{L}_{N\to A}. \tag{12}$$

To satisfy the MIL assumption, only the most informative snippets are selected using top-K pooling:

$$K = \left\lfloor \frac{T}{16} \right\rfloor + 1. \tag{13}$$

Top-K snippets are used for both classification and metric learning, ensuring sparsity and preventing anomaly dilution.

Under weak supervision, only video-level labels $y \in \{0, 1\}$ are available, while frame-level annotations are unknown. Let

$$\mathbf{s} = \{s_t\}_{t=1}^{T}, \quad s_t \in [0, 1], \tag{14}$$

denote the predicted anomaly score for the t-th temporal snippet.

Following the Multiple Instance Learning (MIL) assumption, snippet scores are aggregated using top-K pooling:

$$\hat{s} = \frac{1}{K} \sum_{t \in \mathrm{TopK}(\mathbf{s})} s_t, \tag{15}$$

where

$$K = \left\lfloor \frac{T}{16} \right\rfloor + 1. \tag{16}$$

The MIL loss is defined using BCE:

$$\mathcal{L}_{\mathrm{MIL}} = -\left[y \log(\hat{s}) + (1 - y) \log(1 - \hat{s}) \right]. \tag{17}$$

This formulation enforces that at least one snippet in an abnormal video receives a high anomaly score, while all snippets in normal videos are encouraged to remain non-anomalous

We apply a triplet margin loss to explicitly separate normal and abnormal representations:

$$\mathcal{L}_{\mathrm{triplet}} = \max\big(0, \|\mathbf{f}_a - \mathbf{f}_p\|_2 - \|\mathbf{f}_a - \mathbf{f}_n\|_2 + m\big), \tag{18}$$

where $\mathbf{f}_a$ denotes a normal anchor, $\mathbf{f}_p$ denotes normal snippets within abnormal videos, and $\mathbf{f}_n$ denotes anomalous snippets. This loss enforces intra-class compactness and inter-class separation.

To handle label ambiguity and feature uncertainty, memory-augmented features are passed through a variational encoder:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}). \tag{19}$$

The variational encoder predicts a mean vector $\boldsymbol{\mu}$ and a standard deviation vector $\boldsymbol{\sigma}$ that parameterize a Gaussian latent distribution for each memory-augmented feature.

The latent distribution is regularized using the KL divergence:

$$\mathcal{L}_{\mathrm{KL}} = -\frac{1}{2} \sum (1 + \log \sigma^2 - \mu^2 - \sigma^2). \tag{20}$$

This encourages smooth latent representations and improves robustness to noisy supervision.

To impose a global embedding structure, we introduce a distance-based margin loss:

$$\mathcal{L}_{\mathrm{dist}} = \max\big(0, M - \|\mathbf{f}_{\mathrm{abn}}\|_2 + \|\mathbf{f}_{\mathrm{norm}}\|_2\big), \tag{21}$$

which enforces abnormal features to reside farther from the origin than normal features. This complements the relative constraints imposed by triplet loss. Here, $\mathbf{f}_{\text{abn}}$ and $\mathbf{f}_{\text{norm}}$ denote the aggregated latent feature representations of abnormal and normal video snippets, respectively. Specifically, $\mathbf{f}_{\text{abn}}$ is obtained by applying top-K pooling over the memory-augmented features corresponding to high-confidence anomalous snippets from abnormal videos, while $\mathbf{f}_{\text{norm}}$ is computed by aggregating features from normal videos using the normal memory branch. Both representations are extracted after temporal attention, dual-memory interaction, and variational encoding, ensuring they capture discriminative and uncertainty-aware characteristics. By enforcing $|\mathbf{f}_{\text{abn}}|_2 > |\mathbf{f}_{\text{norm}}|_2$, the proposed distance-based loss establishes a global ordering in the embedding space, encouraging abnormal patterns to occupy regions farther from the origin and thereby complementing the relative separation enforced by the triplet loss.

Frame-level predictions in WSAD often suffer from temporal jitter. To address this, we introduce a temporal smoothness loss:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{T-1} \sum_{t=1}^{T-1} \left| s_{t+1} - s_t \right|, \tag{22}$$

where s_t denotes the predicted anomaly score at time t. This loss encourages temporally coherent predictions while preserving sharp anomaly boundaries due to its low weighting.

2.2 Overall Training Objective

The final loss function is a weighted combination:

$$\begin{aligned} \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MIL}} + \lambda_1 \mathcal{L}_{\text{memory}} + \lambda_2 \mathcal{L}_{\text{triplet}} \\ + \lambda_3 \mathcal{L}_{\text{KL}} + \lambda_4 \mathcal{L}_{\text{dist}} + \lambda_5 \mathcal{L}_{\text{smooth}}. \end{aligned} \tag{23}$$

Each term addresses a distinct failure mode of weakly supervised anomaly detection.

To improve generalization to unseen anomaly patterns, we adopt (MAML). The general aspects of MAML are given in Zhang et al. (2022); Naik et al. (2025). Each meta-iteration consists of a support set $\mathcal{S}$ and a query set $\mathcal{Q}$. The phase where the model is trained over $\mathcal{S}$ is referred to as 'pseudo-training' and the query validation over $\mathcal{Q}$ is referred to as 'pseudo-testing'.

The model adapts to $\mathcal{S}$ via one gradient step:

$$\boldsymbol{\theta}' = \boldsymbol{\theta} - \alpha \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{S}}. \tag{24}$$

where $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{S}}$ is the gradient of the pseudo-training loss $\mathcal{L}_{\mathcal{S}}$ with respect to model parameters $\boldsymbol{\theta}$.

The adapted parameters are evaluated on $\mathcal{Q}$:

$$\mathcal{L}_{\text{meta}} = \mathcal{L}_{\mathcal{S}} + \beta \mathcal{L}_{\mathcal{Q}}, \tag{25}$$

while gradients are backpropagated through the adaptation step. $\mathcal{L}_{\mathcal{S}}$ and $\mathcal{L}_{\mathcal{Q}}$ are the losses in pseudo-train and pseudo-test phases.

At the beginning, we introduce a warmup phase without any meta-learning for a small number of steps. This helps startup the memory representations without any noisy gradient updates that could result from meta-learning. After the MAML steps begin, we perform second-order gradient updates through both pseudo-train and pseudo-test adaptations, enabling true meta-learning. After full training, a validation step outputs frame-level anomaly scores.

2.3 UCF dataset

The UCF dataset, described in Soomro (2012), is a comprehensive large dataset with a wide variety of normal and abnormal videos. The training set contains 800 normal videos and 810 abnormal videos while the testing dataset has 290 videos (150 normal and 140 abnormal). There are 13 different classes of anomalies such as Abuse, assault, arson, arrest, shoplifting, vandalism, robbery, burglary, explosion, fighting, road accidents, shooting, stealing and thus constitutes a robust and challenging dataset for anomaly detection.

3 Results

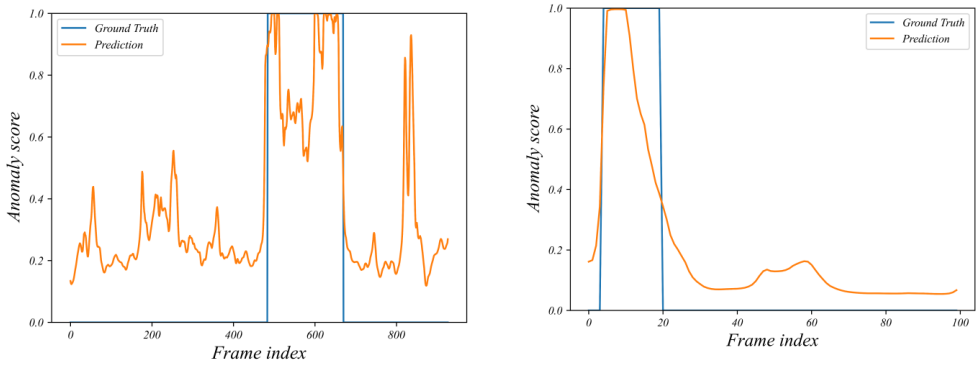
The proposed model was evaluated on a benchmark UCF-Crime dataset. After significant training, the model was evaluated on the test set to obtain frame level anomaly scores ranging from 0 to 1. A value of 0.5 and above was the threshold for anomaly classification. We achieve a frame-level AUC of 86% and an overall accuracy of 87%. This shows that the proposed model can reliably localize anomalous segments within long, untrimmed videos. This is a strong result for a WSAD setup since the model has to learn to capture anomalies which are present in only a small number of frames with the rest of the frames being normal. The robust overall accuracy scores also show that the model generalizes well across diverse anomalies and prevents overfitting. This is a very common limitation in earlier WSAD systems, since it is easy to learn frequent motion patterns. Thus our model exceeds the typical performance range of classical MIL-based WSAD methods and highlights the effectiveness of incorporating meta-learning adaptation to global temporal modeling with transformer based architectures.

Figure 2 and Figure 3 indicates four representative abnormal video samples for which the proposed method demonstrated strong predictions. In the first example from Figure 2a and Figure 2c , the ground truth(blue line) shows normal activity until approximately frame 450, followed by an anomalous section which continues till frame 650. The predicted scores represented by orange curve is below the threshold level during the normal segment i.e till frame 450, and rising above it during the anomalous interval, from frame 450 to 650 approximately. Although minor fluctuations are observed after frame 650, the overall prediction remains consistent with the ground truth.

The second abnormal case in Figure 2b and Figure 2d demonstrates considerably strong temporal alignment as the ground truth curve indicate abnormal activity during



(a) (b)

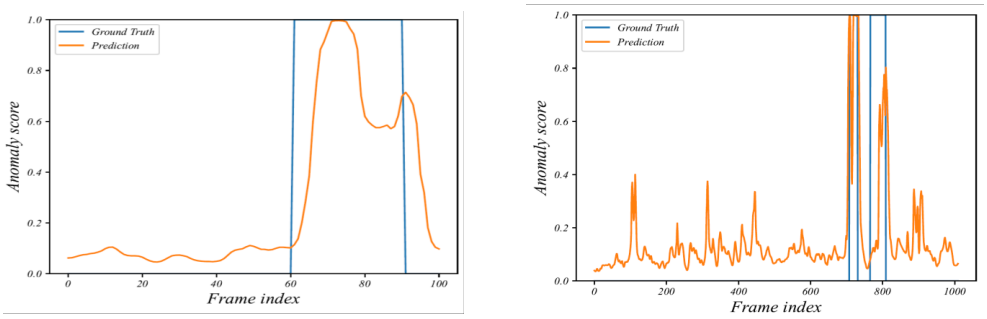


(c) (d)

Fig. 2: Frame level score comparison of abnormal videos between ground truth (blue) and prediction (orange). (a) Example frames from a burglary scenario, where the upper red bounding box highlights the anomalous region, while the lower green box denotes concurrent normal activity. (c) Corresponding frame-level anomaly score plot for the burglary video over frames 450 to 650, demonstrating close alignment between the predicted anomaly scores and the ground-truth annotations. (b) Example frames from a shooting scenario, in which the upper green box indicates normal behaviour and the lower red box marks the shooting event. (d) The associated anomaly score plot, showing a clear anomaly response within the frame interval 0 to 20, consistent with the annotated abnormal event.



(a) (b)



(c) (d)

Fig. 3: Frame level score comparison of abnormal videos between ground truth (blue) and prediction (orange). (a) Example frames from a explosion scenario, where the upper red bounding box highlights the anomalous region, while the lower green box denotes concurrent normal activity. (c) Corresponding frame-level anomaly score plot for the explosion video over frames 60 to 90, demonstrating close alignment between the predicted anomaly scores and the ground-truth annotations. (b) Example frames from an assault scenario, in which the upper green box indicates normal behaviour and the lower red box marks the shooting event. (d) The associated anomaly score plot, showing a clear anomaly response within the frame interval 750 to 800 with ups and downs, consistent with the annotated abnormal event.

the first 20 frames as the scores are above threshold, which is aligned with model's predictions very well. From frame 20 onward, both the ground truth and predicted scores are falling below threshold levels indicating a normal activity.

In the third example in Figure 3a and Figure 3c, frame 0 to 60 a normal activity is shown by ground truth, followed by a brief anomalous segment between frames 60 and 90, and normal behavior thereafter. The predicted anomaly scores successfully mirror this temporal structure, correctly identifying both the onset and duration of the abnormal event.

Finally, the fourth example in Figure 3b and Figure 3d shows normal activity during the initial 100 frames, followed by a short anomalous interval between frames 100 and 120. The model identified anomalous section clearly as indicated by scores which are above threshold. Also it could predict normal patterns in the initial time frame matching the ground truth annotations.

Figure 4 presents qualitative comparisons between the predicted anomaly scores (orange curves) and the ground-truth frame-level annotations (blue curves) for both normal and abnormal video samples. These plots enable a detailed inspection of the model's temporal localization performance under diverse scenarios.

Figures 4a and 4c correspond to normal video sequences exhibiting contrasting prediction behaviors. In the first case, predicted scores from the model are consistently below the threshold of 0.5 which indicates that it is purely a normal video. Whereas in second example, Figure 4b and Figure 4d, the model falsely predicts anomaly in the initial 300 frames, indicating abnormal activity in contrast to ground truth which says the video is a normal activity.

Analysis of these scores confirm that the model can effectively distinguish normal and abnormal frames and thus can be very effective in real world scenarios.

Table 1: Quantitative Results on UCF-Crime

Method	Frame AUC (%)	Accuracy (%)
Sultani et al. (2018)	75.4	78.2
MIST (2020)	82.3	83.1
RTFM (2021)	83.1	84.0
Anomaly Transformer (2021)	82.7	83.5
Proposed method	**86**	**87.0**

3.1 Ablation Studies

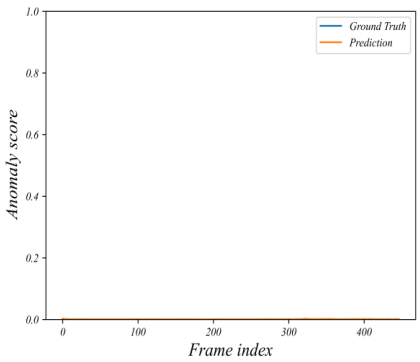
We conducted comprehensive ablation studies to assess the contribution of each major component in the proposed meta-learning based UR-DMU framework. We observed that the absence of dual memory unit significantly degrades performance, emphasizing the importance of separating normal and abnormal representations for robust anomaly discrimination. Removal of transformer-based temporal modeling module reduces detection accuracy, particularly for long or gradually evolving anomalies, as it is unable to capture global temporal context. without integrating variational



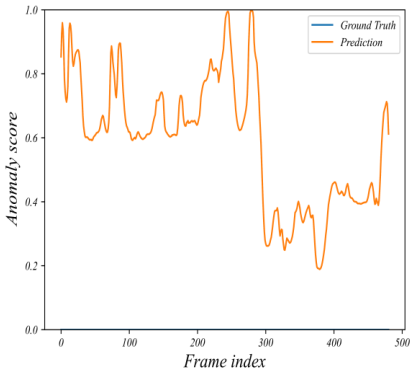
(a)



(b)



(c)



(d)

Fig. 4: Frame level score comparison of normal videos between ground truth (blue) and prediction (orange). (a) and (c) show a screen snip and anomaly score plots respectively, for a good matching video. The scores in (c) are zero for both ground truth and prediction. (b) and (d) are the same for a bad result, where the prediction is abnormal (threshold greater than 0.5 score) for upto 300 frames.

autoencoder it was observed that the model exhibits faster convergence but poorer generalization, indicating that latent space regularization is crucial for stable task adaptation in the MAML setting. Also if local attention mechanism is disabled, it leads to less precise temporal localization, as the model fails to capture fine-grained anomaly

cues. Overall, these ablations demonstrate that each component contributes complementary benefits, and their combined integration is essential for achieving strong performance in weakly supervised video anomaly detection.

Table 2: Ablation studies on model architecture

Model components	AUC (%)	Accuracy (%)
No dual memory	81.22	82
No transformer	80.9	83.1
No VAE	82.76	84.0
No local attention	80.18	85.8

4 Discussion and Conclusion

Our work implemented a meta-learning-driven framework for weakly supervised video anomaly detection that addresses limitations of conventional MIL-based WSAD approaches, which suffer from poor generalization over unseen anomaly patterns. This is because in abnormal videos, anomalies are usually of very short duration, while the rest of the frames are normal. In the absence of meta-learning, a weakly supervised approach would need heavy training to achieve robust frame-level localization of anomalies. By reformulating anomaly detection as a task-adaptive learning problem, the model enables effective handling of task-level variability.

We have thus demonstrated a powerful proof-of-concept using only video-level supervision which generalizes over unseen patterns. Usage of convolutional or recurrent architectures alone is insufficient to capture long-term dependencies present in complex videos. The key novelty of our approach is successful interfacing of transformer-based temporal attention mechanism with local attention and dual memory architecture. Metric learning with latent uncertainty regularization enhances WSAD as shown by our results.

The model combines unique advantages of both meta-learning and memory based methods. With MAML, the model learns task-level patterns rapidly rather than fixed, global representations. This is achieved by splitting of the dataset into pseudo-train and pseudo-test sets. The memory banks further stabilize this process by explicitly recording normal and abnormal patterns. Thus, both MAML and memory units reinforce training which has the added benefits of faster convergence and reduced oscillations in gradients. While traditional MIL methods can overfit or do not generalize well, MAML refines noisy MIL sections through inner-loop updates. Additionally, variational learning is combined with metric based constraints to separate normal and abnormal representations at the feature level.

Future work will also explore more efficient meta-training strategies, adaptive task sampling, and tighter integration of reconstruction-based objectives to further enhance robustness under open-set and low-resource conditions. Minimizing the computational

overhead of meta-learning will be a key area of focus. Overall, this work represents a novel approach toward adaptive, scalable, and deployment ready video anomaly detection systems, for real-world surveillance requirements.

References

Amrutha CV, Jyotsna C, Amudha J (2020) Deep learning approach for suspicious activity detection from surveillance video. In: 2nd International Conference on Innovative Mechanisms for Industry Applications (ICIMIA), IEEE

Bertasius G, Wang H, Torresani L (2021) Is space-time attention all you need for video understanding? In: Proceedings of the International Conference on Machine Learning (ICML)

Chang H, Li Y, Wang X (2020) Data uncertainty learning for feature embeddings in face recognition. IEEE Transactions on Pattern Analysis and Machine Intelligence 42(10):2483–2495

Dilek E, Dener M (2025) An overview of transformers for video anomaly detection. Neural Computing and Applications pp 1–33

Direkoglu C (2020) Abnormal crowd behavior detection using motion information images and convolutional neural networks. IEEE Access 8:80408–80416

Dosovitskiy A, Beyer L, Kolesnikov A, et al (2021) An image is worth 16x16 words: Transformers for image recognition at scale. In: International Conference on Learning Representations (ICLR)

Feng JC, Hong FT, Zheng WS (2021) Mist: Multiple instance self-training framework for video anomaly detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp 14009–14018

Gandapur MQ, Verdú E (2023) Convgru-cnn: Spatiotemporal deep learning for real-world anomaly detection in video surveillance systems. arXiv preprint

Gong D, Liu L, Le V, et al (2019) Memorizing normality to detect anomaly: Memory-augmented deep autoencoder for unsupervised anomaly detection. Proceedings of the IEEE International Conference on Computer Vision pp 1705–1714

Habeb MH, Salama M, Elrefaei LA (2024) Enhancing video anomaly detection using a transformer spatiotemporal attention unsupervised framework for large datasets. Algorithms 17(7):286

Ilse M, Tomczak J, Welling M (2018) Attention-based deep multiple instance learning. In: International conference on machine learning, PMLR, pp 2127–2136

Kanishkar AV, Nithesh B, Nithish Kumar R, et al (2024) Video anomaly detection using liquid neural networks. In: Proceedings of the International Conference on Inventive Communication and Computational Technologies (ICICCT 2024). Springer, pp 159–171, first Online: 15 December 2024

Lee J, Kim H, Park S (2021) Uncertainty-aware temporal action localization. Proceedings of the AAAI Conference on Artificial Intelligence 35(10):8690–8698

Liu Z, Lin Y, Cao Y, et al (2021) Swin transformer: Hierarchical vision transformer using shifted windows. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV)

Luo W, Xu W, Tian Y (2021) Action unit memory networks for weakly supervised temporal localization. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp 12345–12355

Naik S, Palaniswamy S, Conci N, et al (2025) Anomaly detection in videos for chain snatching using meta-learning. In: International Conference on Smart Trends for Information Technology and Computer Communications, Springer, pp 13–23

Roy D, et al (2018) Snatch theft detection in unconstrained surveillance videos using action attribute modelling. Pattern Recognition Letters 108:56–61

Shao W, Rajapaksha P, Wei Y, et al (2023) Covad: Content-oriented video anomaly detection using a self-attention-based deep learning model. Virtual Reality & Intelligent Hardware 5(1):24–41

Soomro K (2012) Ucf101: A dataset of 101 human actions classes from videos in the wild. arXiv preprint arXiv:12120402

Stephen J, Menon A (2020) Readable and updatable memory-augmented autoencoders for video anomaly detection. arXiv preprint arXiv:200301040

Sultani W, Chen C, Shah M (2018) Real-world anomaly detection in surveillance videos. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp 6479–6488

Tian Y, Pang G, Chen Y, et al (2021) Weakly supervised video anomaly detection with robust temporal feature magnitude learning. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp 4975–4986

Vaswani A, Shazeer N, Parmar N, et al (2017) Attention is all you need. In: Advances in Neural Information Processing Systems (NeurIPS)

Viji S, Kannan R, Yogambal Jayalashmi N (2024) Intelligent anomaly detection model for atm booth surveillance using machine learning algorithm: Intelligent atm surveillance model. In: Recent Trends in Artificial Intelligence and Smart Systems.

Springer, https://doi.org/10.1007/978-981-97-7710-5_13

Wang X, Yan Y, Tang P, et al (2018) Revisiting multiple instance neural networks. Pattern recognition 74:15–24

Wu J, Zhang W, Li G, et al (2021) Weakly-supervised spatio-temporal anomaly detection in surveillance video. In: Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)

Wu Z, et al (2022) Ur-dmu: Uncertainty-regularized dual memory unit for weakly supervised video anomaly detection. In: ECCV

Zhang C, Li G, Xu Q, et al (2022) Weakly supervised anomaly detection in videos considering the openness of events. IEEE transactions on intelligent transportation systems 23(11):21687–21699

Zhou H, Yu J, Yang W (2023) Dual memory units with uncertainty regulation for weakly supervised video anomaly detection. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp 3769–3777